

Rift Valley Lake Early Warning System: A Stochastic Control Theory Framework

Combining Hydrological Prediction with Optimal Decision-Making
Under Uncertainty

Macharia Bari

`bariimacharia@gmail.com`

Based on Stochastic Control Theory [1] and the Rift Valley Lakes model [2]

August 2026

Abstract

This document specifies an early warning system for the Kenyan Rift Valley lakes that combines a hydrological water balance model with the Stochastic Control Theory (SCT) framework for universal decision-making under uncertainty. The system treats lake level management as a stochastic control problem: the state is the lake level, the actions are warning and evacuation decisions, and the objective is to minimize flood damage while avoiding false alarms. We define the complete state space, action space, sensing instruments, data pipeline, decision outputs, and the optimal threshold policy derived from the Hamilton-Jacobi-Bellman equation. The system provides 3–6 month lead times with quantified uncertainty, enabling automated warnings and human-in-the-loop evacuation decisions.

1 System Overview

The early warning system has three layers:

1. **Sensing Layer:** Physical instruments and satellite systems that collect raw data
2. **Estimation Layer:** A Kalman filter that fuses sensor data into real-time state estimates with uncertainty quantification
3. **Decision Layer:** An SCT-based optimal control policy that determines when to issue warnings and trigger evacuations

The system maps directly onto the stochastic automaton:

$$\mathbf{A} = (S, \mathcal{A}, P, R, \mathcal{C}, \mathcal{I}, G) \tag{1}$$

Each component of the automaton corresponds to a concrete element of the lake warning problem, as specified below.

2 The Stochastic Automaton for Lake Warning

2.1 State Space $\mathcal{S}$

Definition 2.1 (Lake State). *The state of lake i at time t is:*

$$s_i(t) = (L_i(t), \dot{L}_i(t), P_i(t), E_i(t), G_i(t)) \in \mathcal{S}_i \subset \mathbb{R}^5 \quad (2)$$

where:

- $L_i(t)$: current lake level (meters above datum)
- $\dot{L}_i(t)$: rate of level change (meters/month)
- $P_i(t)$: recent precipitation rate (meters/month)
- $E_i(t)$: recent evaporation rate (meters/month)
- $G_i(t)$: estimated groundwater inflow (meters/month)

The full system state for all eight lakes is:

$$s(t) = (s_{Turkana}(t), s_{Baringo}(t), \dots, s_{Solai}(t)) \in \mathcal{S} = \prod_{i=1}^8 \mathcal{S}_i \quad (3)$$

2.2 Action Space $\mathcal{A}$

Definition 2.2 (Warning Actions). *For each lake i , the action at time t is:*

$$a_i(t) \in \mathcal{A}_i = \{0, 1, 2, 3\} \quad (4)$$

where:

- $a_i = 0$: **Monitor**—no action, continue observation
- $a_i = 1$: **Alert**—issue public warning, prepare evacuation plans
- $a_i = 2$: **Evacuate**—order evacuation of flood-prone areas
- $a_i = 3$: **Emergency**—mandatory evacuation, deploy emergency services

The full action vector is $a(t) = (a_1(t), \dots, a_8(t)) \in \mathcal{A} = \prod_{i=1}^8 \mathcal{A}_i$.

2.3 Transition Kernel P

The lake level evolves according to the water balance model [2]:

$$L_i(t+1) = \eta_i L_i(t) + \alpha_i (P_i(t) - \beta_i E_i(t)) + \gamma_i R_i(t) + \delta_i G_i(t) - \epsilon_i S_i(t) - \zeta_i H_i(t) + \xi_i(t) + w_i(t) \quad (5)$$

where $w_i(t) \sim \mathcal{N}(0, \sigma_{w,i}^2)$ is process noise representing unmodeled dynamics. The transition kernel is:

$$P(s_i(t+1) \mid s_i(t), a_i(t)) = \mathcal{N}(\mu_i(s_i(t)), \Sigma_i(s_i(t))) \quad (6)$$

where μ_i is the deterministic prediction from Eq. (5) and Σ_i captures combined process and observation uncertainty. The action $a_i(t)$ does not affect the lake dynamics directly (the lake does not respond to warnings), but affects the *payoff* through damage mitigation.

2.4 Payoff Structure R

Definition 2.3 (Payoff Function). *The payoff for lake i at time t is:*

$$R_i(s_i, a_i, s'_i) = -C_i^{\text{flood}}(s'_i) \cdot \mathbb{1}\{L'_i > L_i^{\text{crit}}\} \cdot m_i(a_i) - C_i^{\text{false}}(a_i) \cdot \mathbb{1}\{L'_i \leq L_i^{\text{crit}}\} \quad (7)$$

where:

- C_i^{flood} : flood damage cost (lives + property) as a function of the new level L'_i
- L_i^{crit} : critical level threshold for flooding
- $m_i(a_i)$: mitigation factor: $m_i(0) = 1$ (no mitigation), $m_i(1) = 0.5$ (alert reduces damage 50%), $m_i(2) = 0.2$ (evacuation reduces 80%), $m_i(3) = 0.05$ (emergency reduces 95%)
- C_i^{false} : false alarm cost (economic loss, public trust erosion) as a function of action severity

The total system payoff is $R(s, a, s') = \sum_{i=1}^8 R_i(s_i, a_i, s'_i)$.

2.5 Constraint Set $\mathcal{C}$

Definition 2.4 (Safety Constraints). *The constraint set encodes two requirements:*

1. **Floor constraint:** Lake level must not exceed the critical flood level without prior warning:

$$L_i(t) \leq L_i^{\text{crit}} \quad \text{or} \quad a_i(t-1) \geq 1 \quad (\text{warning issued before flooding}) \quad (8)$$

2. **Target constraint:** False alarm rate must remain below a threshold:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{1}\{a_i(t) \geq 1 \text{ and } L_i(t+1) \leq L_i^{\text{crit}}\} \leq \alpha_{FA} \quad (9)$$

where α_{FA} is the maximum tolerable false alarm rate (e.g., 5%).

2.6 Information Set $\mathcal{I}$

Definition 2.5 (Information Set). *At decision time t , the information available is:*

$$\mathcal{I}_t = \{\hat{s}_i(\tau), P_i(\tau), E_i(\tau), \text{forecast}_i(\tau) : \tau \leq t, i = 1, \dots, 8\} \quad (10)$$

consisting of:

- *Filtered state estimates $\hat{s}_i(\tau)$ from the Kalman filter (posterior mean and covariance)*
- *Historical and real-time meteorological data*
- *Numerical weather prediction forecasts for the next 1–6 months*
- *Model parameter estimates $\hat{\Theta}_i = (\alpha_i, \beta_i, \gamma_i, \delta_i, \epsilon_i, \zeta_i, \eta_i)$ with uncertainty*

2.7 Gödelian Operator G

Definition 2.6 (Gödelian Operator for Lake Warning). *The Gödelian Operator G monitors the system’s own prediction accuracy and adjusts the decision policy accordingly:*

$$G : (\hat{\Theta}, \theta_{act}, \mathcal{I}) \mapsto (\hat{\Theta}', \theta'_{act}, \mathcal{I}') \quad (11)$$

Specifically, G performs three functions:

1. **Model self-check:** *Compare predicted vs. observed lake levels over the last n months. If the mean prediction error exceeds a threshold, flag the model for re-calibration.*
2. **Threshold adaptation:** *If the false alarm rate has been too high, raise the warning threshold θ_{act} . If warnings have been issued too late, lower it.*
3. **Parameter refresh:** *Re-estimate model coefficients $\hat{\Theta}_i$ using the latest data via Bayesian updating.*

3 System Inputs: Sensing and Data Collection

3.1 Primary Sensing Instruments

3.2 Automated Data Pipeline

1. **Satellite ingestion:** Automated download of Sentinel-3 altimetry, Landsat/Sentinel-2 imagery, CHIRPS precipitation, GPM rainfall, SMAP soil moisture, and MODIS thermal data via APIs (Google Earth Engine, Copernicus Open Access Hub, NASA Earthdata).

Parameter	Instrument	Spatial Res.	Temporal Res.	Data Source
Lake level L_i	Radar altimetry (Sentinel-3, Jason-3)	1 lake-wide	Every 10 days	ESA Copernicus / NASA PO-DAAC
Lake level L_i (in-situ)	Pressure transducer + staff gauge	Point measurements	Hourly	Kenya WRMA stations
Precipitation P_i	Rain gauge + CHIRPS satellite	5 km grid	Daily	Kenya Met Dept / UCSB CHIRPS
Evaporation E_i	Pan evaporation + MODIS thermal	Lake surface	Monthly / Daily	Kenya Met Dept / NASA MODIS
Surface area	Landsat-8/9, Sentinel-2	30 m / 10 m	Every 16 days	USGS / ESA
Groundwater G_i	Piezometer network	Point measurements	Monthly	Kenya WRMA / Becht et al.
Runoff R_i	Stream gauges + GPM satellite	River mouth	Daily / Monthly	Kenya WRMA / NASA GPM
Soil moisture	SMAP satellite	9 km grid	Every 2–3 days	NASA SMAP
Temperature	Weather station network	Point measurements	Hourly	Kenya Met Department
Land use / land cover	Sentinel-2, Landsat	10–30 m	Annual	Kenya Land Ministry

Table 1: Sensing instruments and data sources for the early warning system.

2. **In-situ telemetry:** Pressure transducers and rain gauges transmit hourly readings via GSM/LoRaWAN to a central server. Data is quality-checked automatically (range tests, spike detection, gap filling).

3. **Data assimilation:** A Kalman filter merges satellite and in-situ observations:

$$\hat{L}_{i,t|t} = \hat{L}_{i,t|t-1} + K_t(Y_t - H_t\hat{L}_{i,t|t-1}) \quad (12)$$

$$P_{t|t} = (I - K_t H_t)P_{t|t-1} \quad (13)$$

where Y_t is the observation vector (satellite + in-situ), H_t is the observation model, and K_t is the Kalman gain. This produces a filtered state estimate with quantified uncertainty $\hat{\sigma}_{L,i}(t) = \sqrt{P_{t|t}}$.

4. **Forecast ingestion:** Numerical weather prediction (NWP) forecasts for precipitation and temperature are ingested from ECMWF or GFS for the next 1–6 months, providing the forcing terms $P_i(t+k)$, $E_i(t+k)$ for $k = 1, \dots, 6$.

4 Decision Outputs

4.1 Optimal Warning Policy

The SCT framework yields a bang-bang optimal policy for each lake:

Theorem 4.1 (Optimal Warning Threshold). *The optimal action for lake i at time t is:*

$$a_i^*(t) = \begin{cases} 0 \text{ (Monitor)} & \text{if } \hat{L}_i(t) < \theta_i^{(1)} \\ 1 \text{ (Alert)} & \text{if } \theta_i^{(1)} \leq \hat{L}_i(t) < \theta_i^{(2)} \\ 2 \text{ (Evacuate)} & \text{if } \theta_i^{(2)} \leq \hat{L}_i(t) < \theta_i^{(3)} \\ 3 \text{ (Emergency)} & \text{if } \hat{L}_i(t) \geq \theta_i^{(3)} \end{cases} \quad (14)$$

where the thresholds $\theta_i^{(k)}$ are determined by the Bellman equation:

$$V_i(t, L) = \max_{a \in \{0,1,2,3\}} \left\{ R_i(L, a) + \mathbb{E}[V_i(t+1, L') \mid L, a] \right\} \quad (15)$$

subject to the constraint that the expected false alarm rate remains below α_{FA} .

Proposition 4.2 (Threshold Computation). *The thresholds are computed by backward induction:*

1. Set terminal condition: $V_i(T, L) = -C_i^{flood}(L) \cdot \mathbb{1}\{L > L_i^{crit}\}$
2. For $t = T-1, T-2, \dots, 0$: compute $V_i(t, L)$ for all L on a grid
3. The optimal thresholds are the crossing points where the optimal action changes:

$$\theta_i^{(k)} = \inf \{ L : a_i^*(t, L) \geq k \} \quad (16)$$

4.2 Output Products

The system generates the following outputs at each decision epoch (monthly):

4.3 Key Derived Quantities

4.3.1 Hitting Time (Time to Critical Level)

From the SCT hitting time equation, the expected number of months until lake i reaches its critical level:

$$\mathbb{E}[\tau_{crit,i}] = \frac{1}{\lambda_i} + \sum_x P(X_i = x) \cdot \mathbb{E}[\tau_{crit,i} - x] \quad (17)$$

where λ_i is the monthly rise rate and X_i is the monthly increment distribution. For Baringo under baseline conditions:

$$\mathbb{E}[\tau_{crit}] \approx \frac{L_i^{crit} - \hat{L}_i(t)}{\lambda_i} \pm \frac{\sigma_{w,i}}{\lambda_i^2} \sqrt{n} \quad (18)$$

Output	Description	Format
Current state estimate	$\hat{L}_i(t) \pm \hat{\sigma}_i(t)$ for each lake	Numeric + time series plot
6-month forecast	$\hat{L}_i(t+k) \pm 2\hat{\sigma}_i(t+k)$, $k = 1, \dots, 6$	Time series with confidence bands
Warning level	$a_i^*(t) \in \{0, 1, 2, 3\}$ for each lake	Color-coded dashboard (green/yellow/orange/red)
Hitting time estimate	Expected months until $L_i = L_i^{\text{crit}}$	$\mathbb{E}[\tau_{\text{crit}}]$ with 90% CI
Violation probability	$P(\exists t \leq T : L_i(t) > L_i^{\text{crit}})$	Probability score per lake
False alarm rate	Rolling 12-month false alarm percentage	Monitor for constraint compliance
Model confidence	Posterior $\hat{\Theta}_i$ with credible intervals	Coefficient table
Gödelian reflection report	Model self-check results, threshold adjustments	Text + diagnostics

Table 2: System output products.

4.3.2 Violation Probability

The probability that the lake exceeds its critical level within the next T months:

$$P(\exists t \leq T : L_i(t) > L_i^{\text{crit}}) \approx 1 - \Phi\left(\frac{L_i^{\text{crit}} - \hat{L}_i(t) - \mu_i T}{\sigma_{w,i} \sqrt{T}}\right) \quad (19)$$

where μ_i is the expected monthly rise and Φ is the standard normal CDF.

4.3.3 Early Warning Lead Time

The lead time is the gap between warning issuance and flooding:

$$\text{Lead time}_i = \mathbb{E}[\tau_{\text{crit},i}] - t_{\text{warning},i} \quad (20)$$

Target: ≥ 3 months for Evacuate, ≥ 6 months for Alert.

5 Numerical Example: Lake Baringo

Using the calibrated coefficients from [2] ($\alpha = 0.80$, $\eta = 0.95$, baseline rise rate $\lambda \approx 0.6$ m/year), and assuming Baringo’s critical flood level is $L^{\text{crit}} = 12.0$ m (6 m above the 2010 baseline of 6.0 m):

Current Level	$\mathbb{E}[\tau_{\text{crit}}]$	Violation Prob.	Action
7.0 m	20 months	0.82	Alert ($a = 1$)
8.0 m	10 months	0.91	Evacuate ($a = 2$)
9.0 m	5 months	0.97	Emergency ($a = 3$)
10.0 m	2 months	0.99	Emergency ($a = 3$)

Table 3: Warning thresholds for Lake Baringo (illustrative).

The system would have issued an Alert for Baringo when levels reached 7.0 m (approximately mid-2018), an Evacuate order at 8.0 m (late 2019), and an Emergency at 9.0 m (early 2020), providing 5–20 months of lead time for each escalation.

6 Gödelian Self-Monitoring

The Gödelian Operator G provides continuous model validation:

1. **Prediction accuracy monitor:** Every month, compute the rolling 12-month mean absolute error (MAE) between predicted and observed lake levels. If $\text{MAE}_i > 0.05$ m for two consecutive months, trigger parameter recalibration.
2. **False alarm rate tracker:** Compute the rolling 12-month false alarm rate. If it exceeds $\alpha_{\text{FA}} = 5\%$, increase all thresholds $\theta_i^{(k)}$ by 0.1 m. If the rate is below 1%, decrease thresholds by 0.05 m to improve sensitivity.
3. **Structural break detection:** Apply a CUSUM test to the prediction residuals. A sustained shift in mean residual indicates a structural change in the lake’s hydrology (e.g., new groundwater pathway, dam construction, land use change) that requires model re-specification, not just re-calibration.
4. **Cross-lake consistency check:** If two adjacent lakes (e.g., Baringo and Bogoria) show divergent trends without an explanatory physical cause, flag both models for review. The Gödelian Limit ensures this self-monitoring is approximate—the system can detect its own errors but cannot guarantee perfection.

7 Implementation Requirements

7.1 Hardware

- Central server (cloud or local) for data assimilation and model computation
- $8 \times$ pressure transducer stations (one per lake, telemetry-enabled)
- $20 \times$ rain gauges (distributed across the eight catchments)
- $8 \times$ stream gauges (at major river inflows)
- $8 \times$ piezometers (groundwater monitoring, one per lake basin)

- Satellite data subscriptions (free via Copernicus and NASA)
- Communication infrastructure: GSM/LoRaWAN for in-situ data, internet for satellite data and alert dispatch

7.2 Software

- Data ingestion pipeline: Python scripts fetching from Google Earth Engine, Copernicus, NASA Earthdata APIs
- Kalman filter: Open-source implementation (e.g., `pykalman` or custom)
- Water balance model: Nelder-Mead calibration (SciPy), forward prediction
- Bellman solver: Value iteration on a discretized level grid, backward induction
- Dashboard: Web-based (e.g., Grafana or custom React app) with color-coded lake status
- Alert dispatch: SMS gateway (Twilio/Africa’s Talking), radio automation, TV crawl integration

7.3 Deployment Timeline

Phase	Activity	Duration
1	Install in-situ sensors, establish telemetry	3 months
2	Deploy data pipeline and Kalman filter	2 months
3	Calibrate water balance model with real-time data	2 months
4	Implement Bellman solver and threshold policy	1 month
5	Build dashboard and alert dispatch	2 months
6	Testing, validation, Gödelian operator tuning	2 months
7	Operational deployment	Ongoing

Table 4: Implementation timeline.

8 Conclusion

This early warning system combines a physically grounded water balance model with the Stochastic Control Theory framework for optimal decision-making under uncertainty. The 7-tuple $(S, \mathcal{A}, P, R, \mathcal{C}, \mathcal{I}, G)$ maps directly onto the lake warning problem: state estimation via Kalman filtering, threshold-based optimal actions via the Bellman equation, and continuous self-monitoring via the Gödelian Operator. The system provides 3–6 month lead times with quantified uncertainty, enabling automated warnings that save lives and protect property in the Kenyan Rift Valley.

References

- [1] Macharia Barii (2026). Stochastic Control Theory: Universal Decision-Making. *arXiv preprint*.
- [2] Macharia Barii (2025). Dynamics and Prediction of Water Levels in the Kenyan Rift Valley Lakes. *Technical report*.
- [3] Akivaga, E., et al. (2021). Assessment of rising water levels of Rift Valley lakes in Kenya. *ResearchGate*.
- [4] Avery, S. (2023). Kenya’s Rift Valley lakes are rising, putting thousands at risk. *The Conversation*.
- [5] Becht, R., et al. (2021). Groundwater links between Kenyan Rift Valley lakes. *Academia.edu*.
- [6] Chorowicz, J. (2005). The East African Rift System. *Journal of African Earth Sciences*, 43(1-3), 379–410.
- [7] IPCC (2021). Climate Change 2021: The Physical Science Basis. *Cambridge University Press*.
- [8] Kenya Government Report (2020). Impacts of rising lake levels in the Rift Valley.
- [9] Nicholson, S. E. (1996). A review of dynamics and climate variability in Eastern Africa. In *The Limnology, Climatology and Paleoclimatology of the Eastern African Lakes*.
- [10] Obando, J. A., et al. (2016). Impact of short-term flooding on livelihoods in the Kenya Rift Valley lakes. *ResearchGate*.
- [11] Olaka, L. A., et al. (2010). The sensitivity of East African rift lakes to climate fluctuations. *Journal of Paleolimnology*, 44, 629–644.
- [12] Owen, R. B., et al. (2004). Swamps, springs and diatoms: Wetlands of the semi-arid Bogoria-Baringo Rift, Kenya. *Hydrobiologia*, 518, 59–78.